

Notes on Lewin Ch4: Spectral Theorem and Functional Calculus

1 Multiplication Operators

Let $B \subset \mathbb{R}^d$ be Borel, and let μ be a locally finite Borel measure on B . We set $\text{frak}(H) = L^2(B, d\mu; \mathbb{C})$. Local finiteness implies that $L_c^\infty \subset \text{frak}(H)$.

Example 1. Let $B = \{x_1, \dots, x_k\} \subset \mathbb{R}^d$ and $\mu = \sum_i \delta_{x_i}$. Then $L^2(B, d\mu) \cong \mathbb{C}^k$, so every operator T identifies under the obvious choice of basis $e_i = \text{bb}(1)_{\{x_i\}}$ with a matrix $M \in \mathbb{C}^{k \times k}$, with $M_{ij} = \langle e_i, T e_j \rangle = (T e_j)(x_i)$. Then $g = T f$ is the element $g(x_i) = \sum_j M_{ij} f(x_j)$.

- A diagonal matrix M corresponds to $f \mapsto a f$ where $a(x_i) = M_{ii}$.
- Every Hermitian matrix $M = M^*$ can be diagonalized (possibly under a different basis).
- Let $a \in L_{\text{loc}}^2(B, d\mu)$, then M_a is the operator defined by

$$M_a f(x) = a(x) f(x), \quad D(M_a) = \{v \in L^2(B, d\mu) : av \in L^2(B, d\mu)\}. \quad (1)$$

Theorem 1. Let $a \in L_{\text{loc}}^2(B, d\mu)$.

1. $(M_a, D(M_a))$ is closed.
2. $\sigma(M_a) = \text{ess ran}(a)$, where the essential range of a is

$$\text{ess ran}(a) = \{y \in \mathbb{C} : \mu(|a(\cdot) - y| \leq \varepsilon) > 0 \text{ for all } \varepsilon > 0\}. \quad (2)$$

1. The eigenvalues of M_a are the $\lambda \in \text{ess ran}(a)$ such that $\mu(a = \lambda) > 0$, with the corresponding eigenspace $L^2(\{a = \lambda\}, d\mu)$, the space of all square-integrable functions with support in the set $\{a = \lambda\}$, defined μ -a.e.
 2. $(M_a, D(M_a))$ is bounded iff $a \in L^\infty(B, d\mu)$.
 3. $(M_a, D(M_a))$ is self-adjoint iff a is real-valued (bounded or not).
- Bits of the proof. For $\lambda \notin \text{ess ran}(a)$, there is $\varepsilon > 0$ such that $|a - \lambda| > \varepsilon$ μ -a.e. Thus, $\frac{1}{a - \lambda} \in L^\infty(B, d\mu)$, and the map $v \mapsto v(a - \lambda)^{-1}$ is bounded $L^2(B, d\mu) \rightarrow D(M_a)$ (using $\frac{a}{a - \lambda} = 1 + \frac{\lambda}{a - \lambda}$) and is an inverse for $M_a - \lambda$. For $\lambda \in \text{ess ran}(a)$, there exists $R_n > 0$ such that $\mu(\{|a - y| < \frac{1}{n}\} \cap \mathbb{B}_{R_n}) \in (0, \infty)$ for all n . Then with $u_n := \text{bb}(1)_{\{|a - y| < \frac{1}{n}\} \cap \mathbb{B}_{R_n}}$ we have

$$\text{norm}((M_a - \lambda)u_n)^2 \leq \frac{1}{n^2} \text{norm}(u_n)^2. \quad (3)$$

So $M_a - \lambda$ cannot be invertible.

Theorem 2 (Theorem 4.4: Spectral Theorem). Let $(A, D(A))$ be self-adjoint on $\text{frak}(H)$. Then there exists $d \geq 1$, a Borel set B , a locally finite measure μ on B , a real-valued locally bounded function $a \in L_{\text{loc}}^\infty(B, d\mu)$, and an isomorphism $U : \text{frak}(H) \rightarrow L^2(B, d\mu)$ such that

$$U A U^{-1} = M_a, \quad U D(A) = D(M_a). \quad (4)$$

One can take $d = 2$, $B = \sigma(A) \times \mathbb{N} \subset \mathbb{R}^2$, $a(s, n) = s$, and μ a finite measure on B .

Corollary (resolvent bound).

$$\text{norm}((A - z)^{-1}) = \frac{1}{d}(z, \sigma(A)). \quad (5)$$

Corollary (isolated eigenvalues). Every isolated point of the spectrum is an eigenvalue.

- Define $f(A) := U^{-1}M_{f(a)}U$, $D(f(A)) = U^{-1}D(M_{f(a)})$ for any such isomorphism. We need to show this is independent of the choice of U .

Theorem 3 (Theorem 4.8: Functional Calculus for bounded Borel functions). Let $(A, D(A))$ be self-adjoint. There exists a unique map

$$f \in \text{scr } (L)^\infty(\mathbb{R}, \mathbb{C}) \mapsto f(A) \in \text{cal } (B)(\text{frak } (H)) \quad (6)$$

defined on the C^* -algebra $\text{scr } (L)^\infty(\mathbb{R}, \mathbb{C})$ of bounded Borel functions on $\mathbb{R}$, with values in the algebra $\text{cal } (B)(\text{frak } (H))$ of bounded operators on $\text{frak } (H)$, such that:

- (i) it is a morphism of C^* -algebras ($\mathbb{C}$ -linear, preserves product and star operation).
- (ii) it is continuous, with norm $\|f(A)\| \leq \sup_{x \in \mathbb{R}} |f(x)|$.
- (iii) if $f(x) = (x - z)^{-1}$ with $z \in \mathbb{C} \setminus \mathbb{R}$, then $f(A) = (A - z)^{-1}$.
- (iv) if $f|_{\sigma(A)} \equiv 0$ then $f(A) = 0$.
- (v) if $|f_n(x)| \leq C$ and $f_n \rightarrow f$ pointwise on $\mathbb{R}$, then $f_n(A)v \rightarrow f(A)v$ for all $v \in \text{frak } (H)$.

Remark 1 (Remark 4.9: Spectral measure). Let v be a unit vector of H . By Theorem 3, the map

$$f \in C_b^0(\mathbb{R}, \mathbb{R}) \mapsto \varphi_v(f) := \text{bangle } (v, f(A)v) \in \mathbb{R} \quad (7)$$

is a continuous linear form. If in addition $f \geq 0$ we can write

$$f(A) = \text{sqrt } (f)(A)^2, \quad (8)$$

which shows that φ_v is a positive linear form on C_b^0 . Hence by Riesz–Markov, there is a unique Borel probability measure $\mu_{A,v}$ on $\mathbb{R}$ such that

$$\text{bangle } (v, f(A)v) = \int_{\mathbb{R}} f(s) \, \mathrm{d}\mu_{A,v}(s). \quad (9)$$

With $(B, \mu) = (\sigma(A) \times \mathbb{N}, \mu)$ and $a(s, n) = s$ from Theorem 2, if $U : H \rightarrow L^2(B, \mu)$ is the corresponding unitary map, we can write

$$\text{bangle } (v, f(A)v) = \text{bangle } (Uv, f(a)Uv)_{L^2(B, \mu)} = \int_B f(a) \, \text{abs } (Uv)^2 \, \mathrm{d}\mu = \int_{\sigma(A) \times \mathbb{N}} f(s) \, \text{abs } (Uv(s, n))^2 \, \mathrm{d}\mu(s, n).$$

Therefore $\mu_{A,v}$ is the pushforward measure¹

$$\mu_{A,v} = a_*(\text{abs } (Uv)^2 \mu), \quad (12)$$

i.e. for every Borel set $E \subset \mathbb{R}$,

$$\mu_{A,v}(E) = \int_{a^{-1}(E)} \text{abs } (Uv(x))^2 \, \mathrm{d}\mu(x) = \int_{E \times \mathbb{N}} \text{abs } (Uv(s, n))^2 \, \mathrm{d}\mu(s, n). \quad (13)$$

In words, $\mu_{A,v}$ is the cylindrical projection on $\sigma(A)$ of the probability measure $|Uv(s, n)|^2 \, \mathrm{d}\mu(s, n)$ on $\sigma(A) \times \mathbb{N}$. We then have $v \in D(A)$ iff $\mu_{A,v}$ has a moment of order two, and in this case

$$\int_{\mathbb{R}} s^2 \, \mathrm{d}\mu_{A,v} = \text{norm } (Av)^2. \quad (14)$$

We also have

$$\int_{\mathbb{R}} s \, \mathrm{d}\mu_{A,v}(s) = \text{bangle } (v, Av). \quad (15)$$

We will in fact need this construction in the proof of Theorem 2.

For every bounded Borel function f , $f(A)$ is defined by Theorem 3. By setting $E_A(B) := \text{bb } (1)_B(A)$ we obtain the spectral theorem in terms of projection-valued measures:

¹The book writes

$$\mathrm{d}\mu_{A,v}(s) = \sum_{n \in \mathbb{N}} |Uv(s, n)|^2 \, \mathrm{d}\mu(s, n) \quad (11)$$

but this is a little fast and loose with notation; the sum is a partial integration of $\mathrm{d}\mu(s, n)$ that comes from the pushforward. The proper way is to define the slice measures $\mu_n(E) = \mu(E \times \{n\})$, then we can write $\mathrm{d}\mu_{A,v}(s) = \sum_{n \in \mathbb{N}} |Uv(s, n)|^2 \, \mathrm{d}\mu_n(s)$.

Theorem 4 (Projection-valued spectral measure). Let $(A, D(A))$ be self-adjoint. There exists a unique map

$$E_A : \text{cal } (B)(\mathbb{R}) \rightarrow \text{cal } (B)(H) \quad (16)$$

from the Borel subsets of $\mathbb{R}$ to the orthogonal projections on H , such that:

(i) $E_A(\emptyset) = 0$ and $E_A(\mathbb{R}) = I$.

(ii) if $(B_n)_n$ are pairwise disjoint Borel subsets of $\mathbb{R}$, then

$$E_A(\cup_n B_n)v = \sum_n E_A(B_n)v \quad (17)$$

for every $v \in H$, where the series converges in H .

(iii) for every pair of Borel subsets $B, C \subset \mathbb{R}$,

$$E_A(B \cap C) = E_A(B)E_A(C). \quad (18)$$

(iv) for every bounded Borel function $f \in \text{scr } (L)^\infty(\mathbb{R}, \mathbb{C})$,

$$f(A) = \int_{\mathbb{R}} f(s) dE_A(s). \quad (19)$$

(v) We can recover

$$A = \int_{\mathbb{R}} s dE_A(s). \quad (20)$$

Definition 1 (Scalar spectral measure). Let v be a unit vector of H . Then $\mu_{A,v}$ and E_A are related by

$$\mu_{A,v}(B) = \text{bangle } (v, E_A(B)v). \quad (21)$$

Corollary (Corollary 4.10: Functional Calculus for locally bounded Borel functions). Let $(A, D(A))$ be self-adjoint and let $f : \mathbb{R} \rightarrow \mathbb{C}$ be a locally bounded Borel function. Then $f(A)$ defined above is independent of the isomorphism U used to represent A as a multiplication operator.

This follows from the functional calculus for bounded functions because we can describe $D(f(A))$ and $f(A)v$ in terms of the corresponding functional calculus for $f_n = f \text{ bb } (1)_{\{|f| < n\}}$,

$$\begin{aligned} v \in D(f(A)) &\iff \limsup_{n \rightarrow \infty} \text{norm } (f_n(A)v) < \infty, \\ f(A)v &= \lim_{n \rightarrow \infty} f_n(A)v. \end{aligned} \quad (22)$$

2 Proof of Theorems 4.4 and 4.8

• Structure of the proof:

1. First we prove the functional calculus for the “resolvent algebra” $\text{cal } (A)$ of resolvents. This gives Theorem 5 using the Stone–Weierstrass theorem. This calculus is for functions in $C_{\text{lim}}^0 := \mathbb{C} + C_0^0(\mathbb{R}, \mathbb{C})$ (continuous functions with equal limits at $\pm\infty$).
2. Deduce Theorem 2.
3. Use the monotone class theorem to deduce uniqueness and hence Theorem 3 (the other properties of Theorem 3 follow directly from the definition).

• C_{lim}^0 is a unital C^* -algebra.

Theorem 5 (Theorem 4.11: Continuous functional calculus). Let $(A, D(A))$ be self-adjoint. There exists a unique map

$$f \in C_{\lim}^0(\mathbb{R}, \mathbb{C}) \mapsto f(A) \in \text{cal}(B)(\text{frak}(H)) \quad (23)$$

such that:

- (i) it is a morphism of C^* -algebras ($\mathbb{C}$ -linear, preserves product and star operation).
- (ii) it is continuous, with $\text{norm}(f(A)) \leq \sup_{x \in \mathbb{R}} \text{abs}(f)$.
- (iii) if $f(x) = (x - z)^{-1}$ with $z \in \mathbb{C} \setminus \mathbb{R}$, then $f(A) = (A - z)^{-1}$.

Proof. Define the resolvent algebra $\text{cal}(A)$ as the algebra generated by constant functions and the rational maps $x \mapsto (x - z)^{-1}$ for $z \in \mathbb{C} \setminus \mathbb{R}$. By the Stone–Weierstrass theorem, $\text{cal}(A)$ is dense in $C_{\lim}^0$. It is clearly a C^* -algebra.

We first show that there is a unique morphism of C^* -algebras $\text{cal}(A) \rightarrow \text{cal}(B)(\text{frak}(H))$ sending $(x - z)^{-1}$ to $(A - z)^{-1}$. We are forced to directly map $(x - z)^{-1}$ to $(A - z)^{-1}$ by (iii) and similarly for linear combinations of products by (i). That this assignment is unique and well-defined follows from

- $((A - z)^{-1})^* = (A - \overline{z})^{-1}$, and
- the resolvent identity

$$(A - z)^{-1} - (A - w)^{-1} = (w - z)(A - z)^{-1}(A - w)^{-1}, \quad (24)$$

which demonstrates commutativity.

For (ii) we will use Lemma 1.

Lemma 1 gives us (ii) by using

$$\text{norm}(f)_{\infty}^2 - |f(x)|^2 \in \text{cal}(A). \quad (25)$$

This continuity allows us to well-define $f(A)$ for all $f \in C_{\lim}^0$ by approximating by elements of $\text{cal}(A)$ (via Stone–Weierstrass) and using the continuity to show that the limit is independent of the choice of approximating sequence. Uniqueness follows from the density of $\text{cal}(A)$ in $C_{\lim}^0$. ■

Lemma 1 (Lemma: Stability under the square root). Let $f \in \text{cal}(A)$ be non-negative on $\mathbb{R}$. Then there is a unique $g \in \text{cal}(A)$ such that $g^2 = f$ and $g \geq 0$ on $\mathbb{R}$. In particular,

$$f(A) = g(A)^2 \geq 0. \quad (26)$$

Proof. Let us write bold letters $\text{bold } (\alpha, p, q, \dots)$ to denote vectors / multiindices of some length and set $(f(\text{bold } (\alpha)))^{\text{bold } (\beta)}$ to mean $\prod_i (f(\alpha_i))^{\beta_i}$ when $\text{bold } (\alpha)$ and $\text{bold } (\beta)$ are the same length.

Note that every $f \in \text{cal } (A)$ can be written as a reduced rational function $f = \frac{P}{Q}$ where P and Q are polynomials with no common roots, and Q has no real roots. Conversely every such rational function is in $\text{cal } (A)$ by decomposing as partial fractions.

Let for $f = \frac{P}{Q} \in \text{cal } (A)$,

$$P = c(x - \text{bold } (\alpha))^{\text{bold } (p)}(x - \text{bold } (z))^{\text{bold } (p)'}, \quad Q = (x - \text{bold } (\xi))^{\text{bold } (q)}, \quad (27)$$

with $\alpha_i \in \mathbb{R}, z_i \in \mathbb{C}, \xi_i \in \mathbb{C}$.

Next suppose that $f \geq 0$ on $\mathbb{R}$. In particular then $f(x) \in \mathbb{R}$ when $x \in \mathbb{R}$, so

$$P \text{ overline } (Q) = Q \text{ overline } (P), \quad (28)$$

which implies that $c \in \mathbb{R}$ by comparing the leading term, and also that

$$(x - \text{bold } (z))^{\text{bold } (p)'}(x - \text{overline } (\text{bold } (\xi)))^{\text{bold } (q)} = (x - \text{bold } (\xi))^{\text{bold } (q)}(x - \text{overline } (\text{bold } (z)))^{\text{bold } (p)'}. \quad (29)$$

Since the fraction $\frac{P}{Q}$ is reduced, none of the z_i can equal the ξ_i , so each $(x - z_i)^{p'_i}$ must be a factor of $(x - \text{overline } (\text{bold } (z)))^{\text{bold } (q)}$. Similarly, each $(x - \text{overline } (\xi_i))^{q_i}$ must be a factor of $(x - \text{bold } (\xi))^{\text{bold } (q)}$. It follows that the complex numbers appear with their conjugates with equal multiplicity, so

$$f(x) = c(x - \text{bold } (\alpha))^{\text{bold } (p)} \text{ abs } \frac{(x - \text{bold } (z))^{\text{bold } (p)'}}{\text{abs}} (x - \text{bold } (\xi))^{\text{bold } (q)}. \quad (30)$$

Now, using that $f \geq 0$ on $\mathbb{R}$, we have $c \geq 0$ and p_i are even. So we can take

$$g(x) = \text{sqrt } (c)(x - \text{bold } (\alpha))^{\text{bold } \frac{p}{2}} \text{ abs } \frac{(x - \text{bold } (z))^{\text{bold } (p)'}}{\text{abs}} (x - \text{bold } (\xi))^{\text{bold } (q)}, \quad (31)$$

as required. ■

Proof (of the Spectral Theorem, Theorem 2). As in Remark 1, we can construct the scalar spectral measure $\mu_{A,v}$ for some unit vector v , as follows. By the Riesz–Markov representation theorem, there is a unique Borel probability measure $\mu_{A,v}$ on $\mathbb{R}$ such that

$$\langle v, f(A)v \rangle = \int_{\mathbb{R}} f(s) d\mu_{A,v}(s) \quad (32)$$

for every bounded continuous function f . Now observe that

$$\langle g(A)v, f(A)v \rangle = \int_{\mathbb{R}} \overline{g(s)} f(s) d\mu_{A,v}(s) \quad (33)$$

so that the map $f \mapsto f(A)v$ is an isometry $C_{\text{lim}}^0(\mathbb{R}, \mathbb{C}) \rightarrow \text{frak}(H)$ with respect to the inner product. By taking the closure, this gives an isometry

$$U : L^2(\mathbb{R}, d\mu_{A,v}) \rightarrow \text{cal}(X)_v := \overline{\{f(A)v : f \in C_{\text{lim}}^0(\mathbb{R}, \mathbb{C})\}}, \quad f \mapsto f(A)v. \quad (34)$$

Note that

$$\langle g(A)v, (A-z)^{-1}f(A)v \rangle = \langle v, (\overline{g})(\bullet - z)^{-1}f(A)v \rangle = \int_{\mathbb{R}} \frac{\overline{g(s)}f(s)}{s-z} d\mu_{A,v}(s) \quad (35)$$

Thus, extending by continuity, we see that $(A-z)^{-1}$ restricted to $\text{cal}(X)_v$ is unitarily equivalent to multiplication by $(s-z)^{-1}$. It follows that A is unitarily equivalent to multiplication by s . Indeed, let

$$B := UAU^{-1} \quad (36)$$

Then $(B-z)^{-1} = U(A-z)^{-1}U^{-1} = M_{(s-z)^{-1}}$. At the same time, from

$$(s-z)(M_s - z)^{-1}f = (M_s - z)(M_s - z)^{-1}f = f \quad (37)$$

we see that $(M_s - z)^{-1} = M_{(s-z)^{-1}}$ as well. In particular the resolvents' ranges are the same, so $D(B) = D(M_s)$, and $I = (B-z)(M_s - z)^{-1}$ gives that $B = M_s$.

If there exists v such that $\text{cal}(X)_v = \text{frak}(H)$, then we are done. If not, we need to iterate the argument.

Lemma 2 (Invariance of $\text{cal}(X)_v$). The subspace $\text{cal}(X)_v$ is invariant under $(A-z)^{-1}$, and $\text{cal}(X)_v^\perp$ is also invariant under $(A-z)^{-1}$, for every $z \in \mathbb{C} \setminus \mathbb{R}$.

Proof (of lemma). $(A-z)^{-1} \text{cal}(X)_v \subset \text{cal}(X)_v$ is because $(A-z)^{-1}f \in C_{\text{lim}}^0(\mathbb{R}, \mathbb{C})$ for every $f \in C_{\text{lim}}^0(\mathbb{R}, \mathbb{C})$, and $(A-z)^{-1}$ is continuous.

And for a similar reason, $(A-z)^{-1} \text{cal}(X)_v^\perp \subset \text{cal}(X)_v^\perp$, as $\langle f(A)v, (A-z)^{-1}w \rangle = \langle ((A - \overline{z})^{-1}f(A)v, w) = \langle ((\bullet - \overline{z})^{-1}f)(A)v, w \rangle = 0$. ■

Now, we can write

$$\text{frak}(H) = \bigoplus_{n \in \mathbb{N}} \text{cal}(X)_{v_n} \quad (38)$$

as follows - let e_n be an orthonormal basis for $\text{frak}(H)$, and put $v_1 = e_1$. Then, let $v_2 = P_{\text{cal}(X)_{v_1}}^\perp e_j$ (orthogonal projection) where e_j is the first basis vector not in $\text{cal}(X)_{v_1}$. Note that $\text{cal}(X)_{v_2} \perp \text{cal}(X)_{v_1}$, since $v_2 \perp \text{cal}(X)_{v_1}$, and

$$\langle f(A)v_1, g(A)v_2 \rangle = \langle v_1, (\overline{f}g)(A)v_2 \rangle = 0. \quad (39)$$

Continuing in this way gives the desired decomposition of $\text{frak}(H)$ into a direct sum of invariant subspaces.

We now have that $(A-z)^{-1}$ on each invariant subspace $\text{cal}(X)_{v_n}$ is unitarily equivalent to multiplication by $(s-z)^{-1}$ on $L^2(\mathbb{R}, d\mu_{A,v_n})$. We can combine them into a single isomorphism to $L^2(B, d\mu)$ with $B = \mathbb{R} \times \mathbb{N}$ and $\mu(V \times \{n\}) = 2^{-n}\mu_{A,v_n}(V)$, and define $a(s, n) = s$. It follows as before that A is unitarily equivalent to multiplication by s . This completes the proof of Theorem 2. $\square$